```
clear; clc; close all;
run(fullfile(fileparts(fileparts(mfilename('fullpath'))),
'setup_project.m'));
D_test=make_deployment(1);
P0=D_test.P0(1:5,:)
complete=make_complete(5)
star=make_star(5)
ring=make_ring(5,1)
for j = 1:200
    r = j/10;
    if isfinite(graph_stats(make_disk(P0, r)).diameter)
        rmin = r;
        break
    end
end
fprintf('r_min = %.1f\n', rmin);
disk=make_disk(P0,rmin)
disk_big=make_disk(P0,rmin+0.4)
Gs_test={complete,star,ring,disk,disk_big};

WAs_test={make_WA(Gs_test{1}),make_WA(Gs_test{2}),make_WA(Gs_test{3}),make_W
A(Gs_test{4}),make_WA(Gs_test{5})};
WBs_test={make_WB(Gs_test{1}),make_WB(Gs_test{2}),make_WB(Gs_test{3}),make_W
B(Gs_test{4}),make_WB(Gs_test{5})};
for i=1:5
    fprintf('Network %d:\n', i);
    disp("Weight Matrix for Network A:")
    disp(WAs_test{i})
    disp("Weight Matrix for Network B:")
    disp(WBs_test{i})
end

for i=1:5
    fprintf('graph_stats, Network %d:\n', i);
    st=graph_stats(Gs_test{i})
end
st_disc=graph_stats(blkdiag(make_complete(3), make_complete(3)))

for i=1:5
    fprintf('Row sums, Network %d (WA, WB):\n', i);
    disp([WAs_test{i}*ones(5,1), WBs_test{i}*ones(5,1)])
end


x0=P0(:,1);
y0=P0(:,2);
K=30;
for i=1:5
    fprintf('Network %d, Protocol A:\n', i);
    XA=simulate_linear(WAs_test{i}, x0, K);
    disp(XA(:, [1:6, end]))
    fprintf('Network %d, Protocol B:\n', i);
```

```
    XB=simulate_linear(WBs_test{i}, x0, K);
    disp(XB(:, [1:6, end]))
end
mean(x0)
dd=Gs_test{4}*ones(5,1);
v=(eye(5)+diag(dd))*ones(5,1);
(v'*x0)/(v'*ones(5,1))
YA=simulate_linear(WAs_test{5}, y0, K);
disp("Spreads at each time step should decrease:")
for i=1:6
    disp(Spread(XA(:,i),YA(:,i)));
end
K=50;
XA=simulate_linear(WAs_test{5}, x0, K);
YA=simulate_linear(WAs_test{5}, y0, K);
disp("Spreads at final time step should be near zero:")
for i=1:5
    disp(Spread(XA(:,end-6+i),YA(:,end-6+i)));
end

g_test=D_test.g
lam=[1; 1; 0; 1; 0.5];
Lam=diag(lam);
ax=x0;
ay=y0;
ax(5)=g_test(1);
ay(5)=g_test(2);
WA=WAs_test{4};
Xa=simulate_anchored(WA, Lam, ax, x0, 6)
Ya=simulate_anchored(WA, Lam, ay, y0, 6)
```

```
P0 =

    0.2652    0.2516
   -2.1646    1.8456
   -1.9007    0.6286
    2.0750   -0.4884
    0.3024   -0.0622


complete =

     0     1     1     1     1
     1     0     1     1     1
     1     1     0     1     1
     1     1     1     0     1
     1     1     1     1     0


star =

     0     1     1     1     1
     1     0     0     0     0
```

```
     1     0     0     0     0
     1     0     0     0     0
     1     0     0     0     0


ring =

     0     1     0     0     1
     1     0     1     0     0
     0     1     0     1     0
     0     0     1     0     1
     1     0     0     1     0

r_min = 2.2

disk =

     0     0     1     1     1
     0     0     1     0     0
     1     1     0     0     0
     1     0     0     0     1
     1     0     0     1     0


disk_big =

     0     0     1     1     1
     0     0     1     0     0
     1     1     0     0     1
     1     0     0     0     1
     1     0     1     1     0

Network 1:
Weight Matrix for Network A:
    0.2000    0.2000    0.2000    0.2000    0.2000
    0.2000    0.2000    0.2000    0.2000    0.2000
    0.2000    0.2000    0.2000    0.2000    0.2000
    0.2000    0.2000    0.2000    0.2000    0.2000
    0.2000    0.2000    0.2000    0.2000    0.2000

Weight Matrix for Network B:
         0    0.2500    0.2500    0.2500    0.2500
    0.2500         0    0.2500    0.2500    0.2500
    0.2500    0.2500         0    0.2500    0.2500
    0.2500    0.2500    0.2500         0    0.2500
    0.2500    0.2500    0.2500    0.2500         0

Network 2:
Weight Matrix for Network A:
    0.2000    0.2000    0.2000    0.2000    0.2000
    0.5000    0.5000         0         0         0
    0.5000         0    0.5000         0         0
    0.5000         0         0    0.5000         0
    0.5000         0         0         0    0.5000
```

```
Weight Matrix for Network B:
         0    0.2500    0.2500    0.2500    0.2500
    0.2500    0.7500         0         0         0
    0.2500         0    0.7500         0         0
    0.2500         0         0    0.7500         0
    0.2500         0         0         0    0.7500

Network 3:
Weight Matrix for Network A:
    0.3333    0.3333         0         0    0.3333
    0.3333    0.3333    0.3333         0         0
         0    0.3333    0.3333    0.3333         0
         0         0    0.3333    0.3333    0.3333
    0.3333         0         0    0.3333    0.3333

Weight Matrix for Network B:
         0    0.5000         0         0    0.5000
    0.5000         0    0.5000         0         0
         0    0.5000         0    0.5000         0
         0         0    0.5000         0    0.5000
    0.5000         0         0    0.5000         0

Network 4:
Weight Matrix for Network A:
    0.2500         0    0.2500    0.2500    0.2500
         0    0.5000    0.5000         0         0
    0.3333    0.3333    0.3333         0         0
    0.3333         0         0    0.3333    0.3333
    0.3333         0         0    0.3333    0.3333

Weight Matrix for Network B:
         0         0    0.3333    0.3333    0.3333
         0    0.6667    0.3333         0         0
    0.3333    0.3333    0.3333         0         0
    0.3333         0         0    0.3333    0.3333
    0.3333         0         0    0.3333    0.3333

Network 5:
Weight Matrix for Network A:
    0.2500         0    0.2500    0.2500    0.2500
         0    0.5000    0.5000         0         0
    0.2500    0.2500    0.2500         0    0.2500
    0.3333         0         0    0.3333    0.3333
    0.2500         0    0.2500    0.2500    0.2500

Weight Matrix for Network B:
         0         0    0.3333    0.3333    0.3333
         0    0.6667    0.3333         0         0
    0.3333    0.3333         0         0    0.3333
    0.3333         0         0    0.3333    0.3333
    0.3333         0    0.3333    0.3333         0

graph_stats, Network 1:
```

```
st =

  struct with fields:

      degree: [5×1 double]
       edges: 10
    diameter: 1

graph_stats, Network 2:

st =

  struct with fields:

      degree: [5×1 double]
       edges: 4
    diameter: 2

graph_stats, Network 3:

st =

  struct with fields:

      degree: [5×1 double]
       edges: 5
    diameter: 2

graph_stats, Network 4:

st =

  struct with fields:

      degree: [5×1 double]
       edges: 5
    diameter: 3

graph_stats, Network 5:

st =

  struct with fields:

      degree: [5×1 double]
       edges: 6
    diameter: 3


st_disc =

  struct with fields:
```

```
      degree: [6×1 double]
       edges: 6
    diameter: Inf

Row sums, Network 1 (WA, WB):
     1     1
     1     1
     1     1
     1     1
     1     1

Row sums, Network 2 (WA, WB):
     1     1
     1     1
     1     1
     1     1
     1     1

Row sums, Network 3 (WA, WB):
     1     1
     1     1
     1     1
     1     1
     1     1

Row sums, Network 4 (WA, WB):
     1     1
     1     1
     1     1
     1     1
     1     1

Row sums, Network 5 (WA, WB):
     1     1
     1     1
     1     1
     1     1
     1     1

Network 1, Protocol A:
    0.2652   -0.2845   -0.2845   -0.2845   -0.2845   -0.2845   -0.2845
   -2.1646   -0.2845   -0.2845   -0.2845   -0.2845   -0.2845   -0.2845
   -1.9007   -0.2845   -0.2845   -0.2845   -0.2845   -0.2845   -0.2845
    2.0750   -0.2845   -0.2845   -0.2845   -0.2845   -0.2845   -0.2845
    0.3024   -0.2845   -0.2845   -0.2845   -0.2845   -0.2845   -0.2845

Network 1, Protocol B:
    0.2652   -0.4220   -0.2502   -0.2931   -0.2824   -0.2851   -0.2845
   -2.1646    0.1855   -0.4020   -0.2552   -0.2919   -0.2827   -0.2845
   -1.9007    0.1195   -0.3855   -0.2593   -0.2908   -0.2830   -0.2845
    2.0750   -0.8744   -0.1371   -0.3214   -0.2753   -0.2868   -0.2845
    0.3024   -0.4313   -0.2479   -0.2937   -0.2822   -0.2851   -0.2845

Network 2, Protocol A:
```

```
    0.2652   -0.2845   -0.1196   -0.1691   -0.1543   -0.1587   -0.1577
   -2.1646   -0.9497   -0.6171   -0.3684   -0.2687   -0.2115   -0.1577
   -1.9007   -0.8178   -0.5512   -0.3354   -0.2522   -0.2032   -0.1577
    2.0750    1.1701    0.4428    0.1616   -0.0038   -0.0790   -0.1577
    0.3024    0.2838   -0.0004   -0.0600   -0.1145   -0.1344   -0.1577

Network 2, Protocol B:
    0.2652   -0.4220   -0.2502   -0.2931   -0.2824   -0.2851   -0.2845
   -2.1646   -1.5571   -1.2733   -1.0175   -0.8364   -0.6979   -0.2848
   -1.9007   -1.3592   -1.1249   -0.9062   -0.7530   -0.6353   -0.2848
    2.0750    1.6226    1.1114    0.7710    0.5050    0.3081   -0.2841
    0.3024    0.2931    0.1143    0.0232   -0.0559   -0.1125   -0.2844

Network 3, Protocol A:
    0.2652   -0.5323   -0.3061   -0.3192   -0.2985   -0.2930   -0.2845
   -2.1646   -1.2667   -0.8208   -0.5724   -0.4401   -0.3684   -0.2845
   -1.9007   -0.6634   -0.5904   -0.4286   -0.3665   -0.3279   -0.2845
    2.0750    0.1589    0.1254   -0.0986   -0.1770   -0.2280   -0.2845
    0.3024    0.8809    0.1691   -0.0038   -0.1406   -0.2054   -0.2845

Network 3, Protocol B:
    0.2652   -0.9311    0.1762   -0.6765    0.0266   -0.5381   -0.2833
   -2.1646   -0.8178   -0.4879   -0.3161   -0.3196   -0.2749   -0.2846
   -1.9007   -0.0448   -0.8085    0.0374   -0.5765   -0.0581   -0.2857
    2.0750   -0.7992    0.5627   -0.8368    0.2034   -0.6666   -0.2826
    0.3024    1.1701   -0.8651    0.3694   -0.7567    0.1150   -0.2865

Network 4, Protocol A:
    0.2652    0.1855    0.1701    0.1076    0.0618    0.0236   -0.1221
   -2.1646   -2.0326   -1.6497   -1.3438   -1.0915   -0.8916   -0.1249
   -1.9007   -1.2667   -1.0380   -0.8392   -0.6918   -0.5738   -0.1239
    2.0750    0.8809    0.6491    0.4894    0.3621    0.2620   -0.1214
    0.3024    0.8809    0.6491    0.4894    0.3621    0.2620   -0.1214

Network 4, Protocol B:
    0.2652    0.1589    0.1650    0.0730    0.0209   -0.0348   -0.2824
   -2.1646   -2.0766   -1.8066   -1.5583   -1.3392   -1.1579   -0.2921
   -1.9007   -1.2667   -1.0615   -0.9010   -0.7954   -0.7046   -0.2882
    2.0750    0.8809    0.6402    0.4818    0.3455    0.2373   -0.2800
    0.3024    0.8809    0.6402    0.4818    0.3455    0.2373   -0.2800

Network 5, Protocol A:
    0.2652    0.1855    0.0943   -0.0070   -0.0717   -0.1160   -0.2021
   -2.1646   -2.0326   -1.4535   -1.0438   -0.7592   -0.5712   -0.2022
   -1.9007   -0.8744   -0.6340   -0.4747   -0.3831   -0.3214   -0.2022
    2.0750    0.8809    0.4173    0.2020    0.0627   -0.0269   -0.2021
    0.3024    0.1855    0.0943   -0.0070   -0.0717   -0.1160   -0.2021

Network 5, Protocol B:
    0.2652    0.1589    0.1650   -0.0086   -0.0587   -0.1341   -0.2845
   -2.1646   -2.0766   -1.5619   -1.2380   -0.9618   -0.7808   -0.2847
   -1.9007   -0.5323   -0.5904   -0.4092   -0.4189   -0.3596   -0.2846
    2.0750    0.8809    0.3954    0.2432    0.0749   -0.0140   -0.2845
    0.3024    0.1465    0.1691   -0.0100   -0.0582   -0.1342   -0.2845
```

```
ans =

   -0.2845


ans =

   -0.1226

Spreads at each time step should decrease:
    3.9397

    2.5329

    1.6789

    1.1138

    0.7370

    0.4878

Spreads at final time step should be near zero:
   3.2690e-08

   2.1630e-08

   1.4313e-08

   9.4705e-09

   6.2666e-09


g_test =

    5.7736    5.3920


Xa =

    0.2652    0.1855    0.6232    0.9515    1.1977    1.3824    1.5209
   -2.1646   -2.0326   -1.9667   -1.9337   -1.9172   -1.9090   -1.9048
   -1.9007   -1.9007   -1.9007   -1.9007   -1.9007   -1.9007   -1.9007
    2.0750    0.8809    1.4645    1.9023    2.2306    2.4768    2.6615
    0.3024    3.3272    3.6191    3.8379    4.0021    4.1252    4.2175


Ya =

    0.2516    0.0824    0.8144    1.3634    1.7751    2.0839    2.3155
    1.8456    1.2371    0.9329    0.7808    0.7047    0.6667    0.6477
```

```
    0.6286    0.6286    0.6286    0.6286    0.6286    0.6286    0.6286
   -0.4884   -0.0997    0.8763    1.6083    2.1572    2.5690    2.8778
   -0.0622    2.6461    3.1341    3.5001    3.7746    3.9805    4.1349
```

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